A SPARKLE input bridge and validation

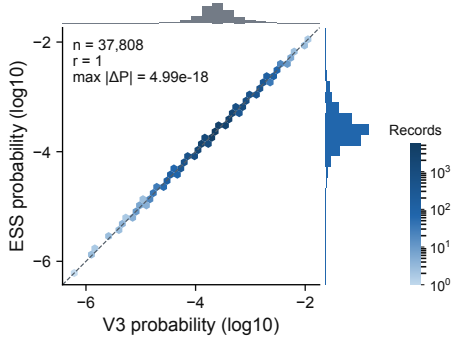
Corrected H5AD → Sparse matrix + metadata → ESS / V3 on identical input

B Bridge and historical regression

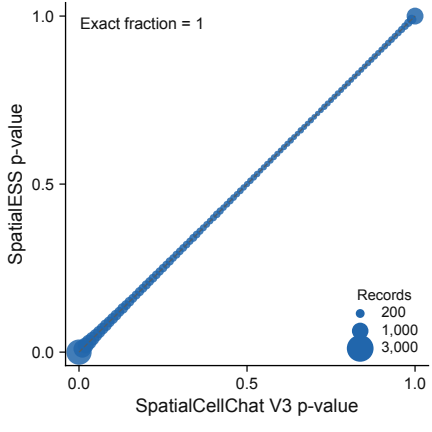
	Max |ΔP|	Exact p	Jaccard	Δ calls
raw 1000 / E	0	1	1	**0**
raw 1000 / B	0	1	1	**0**
raw 5000 / E	0	1	1	**0**
raw 5000 / B	0	1	1	**0**
raw 16247 / E	0	1	1	**0**
raw 16247 / B	0	1	1	**0**
SPK 1000 / E	0	1	1	**0**
SPK 1000 / B	0	1	1	**0**
SPK 5000 / E	0	1	1	**0**
SPK 5000 / B	0	1	1	**0**
SPK 16247 / E	0	1	1	**0**
SPK 16247 / B	0	1	1	**0**

E: historical engine comparison; B: bridge / direct

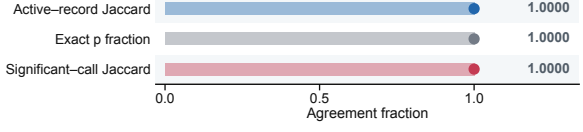
C Corrected-input probabilities



D Permutation p-values



E Interaction-call agreement



F RAW / corrected input sensitivity

